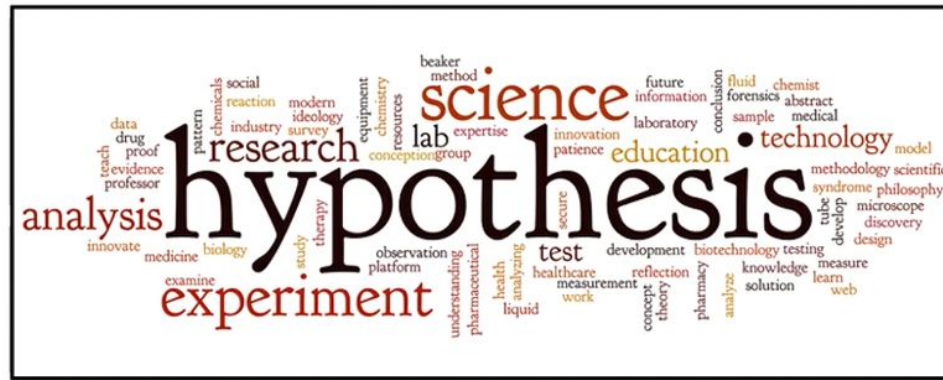


w3.1- One Sample Tests (10:55)

Module 3 - Lesson 1



Let's start by discussing
tests where we compare



Z test

$$H_0: \mu = \mu_0$$

$$H_0: \mu \leq \mu_0$$

$$H_0: \mu \geq \mu_0$$

$$H_1: \mu \neq \mu_0$$

$$H_1: \mu > \mu_0$$

$$H_1: \mu < \mu_0$$

Test statistic:
$$Z = \frac{\bar{X} - \mu_0}{\sigma_{\bar{X}}} = \frac{\bar{X} - \mu_0}{\sigma_x / \sqrt{n}}$$

the population standard
deviation is known,



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Z Test Example

10.20", 8.40", 8.92", 8.87", 10.04", 9.79", 9.84", and 8.71"

$$\sigma_x = 0.35$$

$$\alpha = 0.05$$

$$H_0: \mu \geq 9.25$$

$$H_1: \mu < 9.25$$

Test stat: 0.3930

Critical stat: 1.645

Fail to reject the null hypothesis

As an example,
consider this set of



T Test

$$H_0: \mu = \mu_0$$

$$H_0: \mu \leq \mu_0$$

$$H_0: \mu \geq \mu_0$$

$$H_1: \mu \neq \mu_0$$

$$H_1: \mu > \mu_0$$

$$H_1: \mu < \mu_0$$

$$\text{Test statistic: } t = \frac{\bar{X} - \mu_0}{s/\sqrt{n}}$$



However, the test stat is

T Test Example

$$H_0: \mu = 34 \quad H_1: \mu \neq 34$$

$$\bar{X}=31.7, s = 2.1, n =20$$

Test stat: -4.898 Critical stat: +/- 2.8609

Reject the null hypothesis



Consider this example: Suppose

Chi Square Test

- Applications

- **Comparing variances**
- Comparing observed and expected frequencies

$$H_0: \sigma_x^2 = \sigma_0^2$$

$$H_0: \sigma_x^2 \leq \sigma_0^2$$

$$H_0: \sigma_x^2 \geq \sigma_0^2$$

$$H_1: \sigma_x^2 \neq \sigma_0^2$$

$$H_1: \sigma_x^2 > \sigma_0^2$$

$$H_1: \sigma_x^2 < \sigma_0^2$$

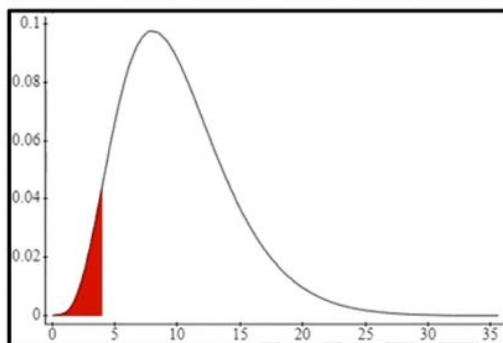
understand the
variance of the data.



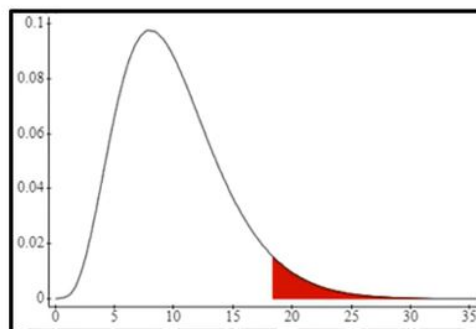
Chi Square Test

Test statistic: $\chi^2 = \frac{(n-1)s^2}{\sigma_x^2}$

Critical statistic: χ^2_α or $\chi^2_{1-\alpha}$
df = n-1



$\chi^2_{1-\alpha}$



χ^2_α

the chi-square test is
given by the formula here.



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Chi Square Test Example

$$\sigma = 20 \quad s = 11.3 \quad n = 10 \quad \alpha = 0.05$$

$$df = 10 - 1 = 9$$

$$\text{Test statistic: } \chi^2 = \frac{(n-1)s^2}{\sigma_1^2} = \frac{(9)(11.3)^2}{(20)^2} = 2.87$$

Critical statistic: 3.33

Reject the null hypothesis



Let's consider this example.

Another Chi Square Application

Leading Digit	1	2	3	4	5	6	7	8	9
Obs. Freq	208	124	106	74	42	84	31	23	17
<u>Benford's Law</u>	30.1%	17.6%	12.5%	9.7%	7.9%	6.7%	5.8%	5.1%	4.6%

Test statistic: $\chi^2 = \sum \frac{(O-E)^2}{E}$ Critical statistic: χ^2_{α} , df = (#rows-1)(#columns -1)



It is often necessary to compare

Benford's Law, also known as the **First-Digit Law**, is a mathematical observation that in many real-life sets of numerical data, the leading digit is not distributed uniformly. 

Contrary to intuition—where you might expect each digit (1 through 9) to appear about 11% of the time—Benford's Law states that the number **1** appears as the leading digit about **30%** of the time, while larger digits like **9** appear much less frequently (less than 5%). 

When does it apply?

Benford's Law works best with data that:

1. **Spans several orders of magnitude** (e.g., populations of all cities in the world).
2. **Is naturally occurring** (not assigned, like phone numbers or ZIP codes).
3. **Has no artificial minimum or maximum** (not something like human height, which stays within a very narrow range). 

Another Chi Square Application (cont.)

Leading Digit	1	2	3	4	5	6	7	8	9
Obs. Freq	208	124	106	74	42	84	31	23	17
<u>Benford's Law</u>	242.004	141.504	100.5	77.988	63.516	53.868	46.632	41.004	36.984

Test statistic = 50.25

$df = (2-1)(9-1) = 8$

Critical statistic = 15.51

Reject the null hypothesis



Proportions Test

- Assumptions
 - Independence
 - **Constant success rate**
 - Two outcomes
 - Finite number of trials
- $np < 5$ and $n(1-p) < 5$ (binomial)
- $np > 5$ and $n(1-p) > 5$ (normal)

When testing a claim about
a population proportion.



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Proportions Test

$$H_0: p = p_0$$

$$H_0: p \leq p_0$$

$$H_0: p \geq p_0$$

$$H_1: p \neq p_0$$

$$H_1: p > p_0$$

$$H_1: p < p_0$$

$$\text{Test statistic: } Z = \frac{x - np_0}{\sqrt{np_0(1 - p_0)}}$$



The configuration of

This transcript provides a comprehensive look at **Parametric Hypothesis Testing** within a Six Sigma Black Belt context. It covers comparing means (Z-tests and T-tests), analyzing variance (Chi-square), checking distribution shapes (Goodness of Fit), and testing proportions.

Below is a structured summary of the key tests and examples discussed.

1. Comparing Means: Z-Tests vs. T-Tests

The choice between these two tests depends primarily on whether the population standard deviation (σ) is known and the size of your sample (n).

Feature	Z-Test	T-Test
Population σ	Known	Unknown
Sample Size	Large ($n > 30$)	Small ($n < 30$)
Distribution	Normal	Student's t (Bell-shaped)
Test Statistic	$Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$	$t = \frac{\bar{x} - \mu}{s / \sqrt{n}}$

Key Insight: If your sample size is over 30, you can use Z by estimating σ with the sample standard deviation (s), but the T-test is more conservative and often preferred in practice for smaller "large" samples.

2. Testing Variance: Chi-Square (χ^2)

This test is used to make inferences about the **population variance**. It is not symmetrical like the Z or T distributions.

- **Primary Uses:** Comparing known variances or comparing observed vs. expected frequencies.
- **Formula:**

$$\chi^2 = \frac{(n - 1)s^2}{\sigma_0^2}$$

- **Degrees of Freedom (df):** $n - 1$

3. Goodness of Fit & Benford's Law

This application of the Chi-square test determines if observed data follows an expected distribution.

- **Example:** Detecting check fraud via **Benford's Law**, which predicts the frequency of leading digits (1 through 9) in natural datasets.
- **The Logic:** If the leading digits of a set of checks deviate significantly from Benford's Law, the null hypothesis is rejected, suggesting the data may be manipulated (fraud).
- **DF for Tables:**

$$(Rows - 1) \times (Columns - 1)$$

4. Proportions Test

Used when the data is **attribute-based (success/failure)** rather than variable-based (measurements).

- **Assumptions:** Independence, constant success rate, and two outcomes (Binomial).
- **Normal Approximation:** You can use the Z-distribution for proportions if **$np > 5$** and $n(1 - p) > 5$.
- **Test Statistic:**

$$Z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

Decision Rule Summary

In almost all these tests, the decision follows the same logic:

- If $|TestStat| > |CriticalStat|$: Reject the Null Hypothesis (H_0).
- If **P-value** $< \alpha$: Reject the Null Hypothesis (H_0).

w3.2- Two Sample Tests (7:27)

Independent Treatment Means Test- Unequal Variances

$$H_0: \mu_1 = \mu_2$$

$$H_1: \mu_1 \neq \mu_2$$

$$\text{Test statistic: } t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Note the hypothesis
statements here are framed



Independent Treatment Means Test- Equal Variances

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

$$t = \frac{\bar{X}_1 - \bar{X}_2}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

$$DF = n_1 + n_2 - 2$$



If we establish the variances

Independent Treatment Example

Freezer 1	Freezer 2
30.7	31.8
31.1	30.1
29.8	28.6
28.8	29.4
32.1	29.5

Equal variances:

$$s_p = 1.511$$

df = 8

Test stat = 0.7975

Critical stat = 3.355

Unequal variances

df = 7.9809

Test stat = 0.7975

Critical stat = 3.358



Let's consider this example.

In statistics, specifically when performing an independent t-test like the one in your image, s_p stands for the **pooled standard deviation**.

Think of it as a "weighted average" of the standard deviations from both groups (Freezer 1 and Freezer 2). We use it when we assume that both groups have **equal variances**, meaning their data spreads out in roughly the same way.

Paired Treatment

$$H_0: \mu_d = 0$$

$$H_1: \mu_d \neq 0$$

$$\text{Test statistic: } t = \frac{\bar{d}}{\left(\frac{s_d}{\sqrt{n}}\right)}$$



We can also compare two groups,

Paired Treatment Example

Before	After
62	65
57	57
53	55
61	63
60	66
66	63
68	70

$$\bar{d} = 1.1714$$

$$s_d = 2.7516$$

Test statistic: 1.6483

$$df = 7 - 1 = 6$$

Critical statistic: 2.445

Fail to reject the null



Let's consider this example.

Difference of Variances Test

$$H_0: \sigma_1^2 = \sigma_2^2$$

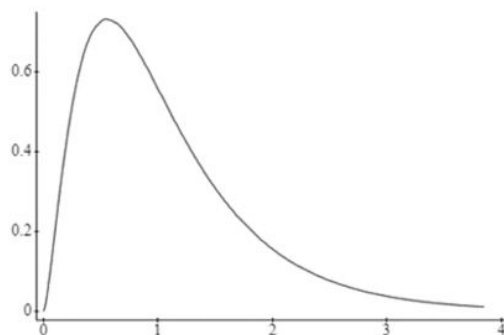
$$H_0: \sigma_1^2 \leq \sigma_2^2$$

$$H_0: \sigma_1^2 \geq \sigma_2^2$$

$$H_1: \sigma_1^2 \neq \sigma_2^2$$

$$H_1: \sigma_1^2 > \sigma_2^2$$

$$H_1: \sigma_1^2 < \sigma_2^2$$



$$\text{Test statistic: } F = \frac{S_1^2}{S_2^2}$$



Homogeneity of variances is

Variance Test Example

	Lot 1	Lot 2
Sample Size	17	11
Standard Deviation	0.248	0.198

$$F = \frac{0.248^2}{0.198^2} = 1.5688$$

$$F_{crit} = 2.233$$

Fail to reject the null



This section covers **Two-Group Parametric Hypothesis Testing**, which is essential for determining if a change in a process (like a freezer setting or a chemical treatment) actually results in a statistically significant difference between two sets of data.

1. Comparing Two Independent Means (2-Sample T-Test)

When comparing the averages of two separate groups (e.g., Freezer A vs. Freezer B), you must first decide if their **variances** are equal.

Equal vs. Unequal Variance

- **Equal Variances:** You "pool" the standard deviations into a single S_p value. This makes the math simpler and the test slightly more powerful.
- **Unequal Variances:** Used when the groups **have different spreads**. The degrees of freedom (df) calculation becomes very complex and is usually handled by software (e.g., Minitab or Excel).

2. Paired Treatment T-Test (Dependent Samples)

This is used when you have a "**Before and After**" scenario on the *same* subjects.

- **How it works:** You don't compare the two group averages directly. Instead, you calculate the **difference** (D) for every single pair and test if the average difference is significantly different from zero.
- **Why use it?** It is more **sensitive** (powerful) because it removes person-to-person or sample-to-sample variation, focusing only on the effect of the treatment.

Example: Measuring material conductivity **before** passivation and **after** passivation on the same 10 samples.

3. Comparing Variances: The F-Test

Before you can trust a T-test, you often need to know if the two groups have the same level of "noise" or spread. We use the **F-test** for this.

- **The Statistic:** It is a simple ratio of the two variances.

$$F = \frac{s_1^2}{s_2^2}$$

- **Distribution:** Unlike the Z or T tests, the F-distribution is **not symmetrical** and starts at zero (since variances can't be negative).
- **Goal:** To check for "Homogeneity of Variances." If F is close to 1, the variances are likely equal.

Summary of Test Statistics

Test Type	Used for...	Key Statistic
Independent T-Test	Two separate groups	Means ($\bar{x}_1, \bar{x}_2$)
Paired T-Test	Before/After on same samples	Mean of differences ($\bar{D}$)
F-Test	Comparing precision/stability	Ratio of variances (s_1^2/s_2^2)

w3.3- Contingency Tables and Proportion Tests (8:19)

Module 3 - Lesson 3



For some, statistical inferences a similar distribution, even if not normal,



Contingency Tables

← Column variable →

↑ Row variable ↓

Contingency Coefficient: $C = \sqrt{\frac{\chi^2}{\chi^2 + N}}$ or $\sqrt{\frac{\chi^2}{N(k-1)}}$ (if $k = \#rows = \#columns$)

$$\text{Max } C = \sqrt{\frac{K-1}{K}} \quad (K = \min \text{ of } (\# \text{ rows}, \# \text{ col.}))$$



Contingency Table Example

Observed Frequencies		Food					Total
		A	B	C	D	E	
Drink	X	13	31	54	27	5	130
	Y	10	45	62	37	41	195
	Z	23	10	8	32	53	126
	Total	46	86	124	96	99	451

Expected Frequencies		Food					Total
		A	B	C	D	E	
Drink	X	13.25942	24.78936	35.74279	27.67184	28.53659	130
	Y	19.88914	37.18404	53.61419	41.50776	42.80488	195
	Z	12.85144	24.02661	34.64302	26.8204	27.65854	126
	Total	46	86	124	96	99	451

$$E = \frac{\text{column total} \times \text{row total}}{\text{grand total}}$$



Contingency Table Example (cont.)

$$\chi^2 = 99.66$$

$$C = .4254$$

$$\text{Max } C = .816$$

$$df = (5-1)(3-1) = 8$$

$$\chi^2_{0.05,8} = 15.51$$

Reject the null hypothesis



Difference of Proportions Test

$$H_0: p_1 = p_2$$

$$H_1: p_1 \neq p_2$$

$$H_0: p_1 \leq p_2$$

$$H_1: p_1 > p_2$$

$$H_0: p_1 \geq p_2$$

$$H_1: p_1 < p_2$$

$$\text{Test statistic: } Z = \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\frac{pq}{n_1} + \frac{pq}{n_2}}}$$

$$\text{Pooled sample proportion: } \bar{p} = \frac{X_1 + X_2}{n_1 + n_2}$$

When finding an inference
about two proportions,



McNemars Test

		Treatment 1	
		Yes	No
Treatment 2	Yes	a	b
	No	c	d

H_0 : The proportions of the frequencies b and c are the same.

H_1 : The proportions of the frequencies b and c are different.

Test statistic:
$$\chi^2 = \frac{(|b - c| - 1)^2}{b + c}$$

When we are dealing with a difference of proportions that are paired or dependent,



This module explores advanced **Chi-Square Applications** and **Proportions Testing**, specifically focusing on how variables interact and how to handle dependent categorical data.

1. Contingency Tables and Dependency

A **contingency table** (or cross-tabulation) is used to determine if two categorical variables are independent or if there is a significant relationship between them.

- **Dependency Test:** Compares "Observed" counts in a table to "Expected" counts. The expected frequency for any cell is calculated as:

$$\text{Expected} = \frac{\text{Row Total} \times \text{Column Total}}{\text{Grand Total}}$$

- **Contingency Coefficient (C):** Measures the strength of the association.

$$C = \sqrt{\frac{\chi^2}{\chi^2 + N}}$$

- **Max Possible C :** Since C is dependent on table size, we compare it to C_{max} to understand its relative strength:

$$C_{max} = \sqrt{\frac{k-1}{k}}$$

(where k is the minimum number of rows or columns).

2. Comparing Two Proportions

This test determines if the success rate of one group is significantly different from another.

- **Assumptions:** Requires at least 5 successes and 5 failures in each group ($np \geq 5$ and $nq \geq 5$).
- **Pooled Proportion ($\bar{p}$):** When the null hypothesis assumes no difference ($P_1 - P_2 = 0$), we combine the data from both groups to create a "pooled" estimate.
- **Test Statistic:**

$$Z = \frac{(\hat{p}_1 - \hat{p}_2) - 0}{\sqrt{\bar{p}\bar{q} \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

3. The McNemar Test (Paired Proportions)

While the standard proportion test is for independent groups, the **McNemar Test** is used for **matched pairs** (e.g., the same person tested twice, or twins).

- **Focus on Discordant Pairs:** The test ignores cases where the results stayed the same (Yes/Yes or No/No). It only analyzes **"Discordant Pairs"**—where the result changed (Yes/No or No/Yes).
- **Formula:**

$$\chi^2 = \frac{(b - c)^2}{b + c}$$

- **Rule of Thumb:** The sum of the discordant pairs ($b + c$) should be at least **10** for the test to be valid.

Comparison Summary

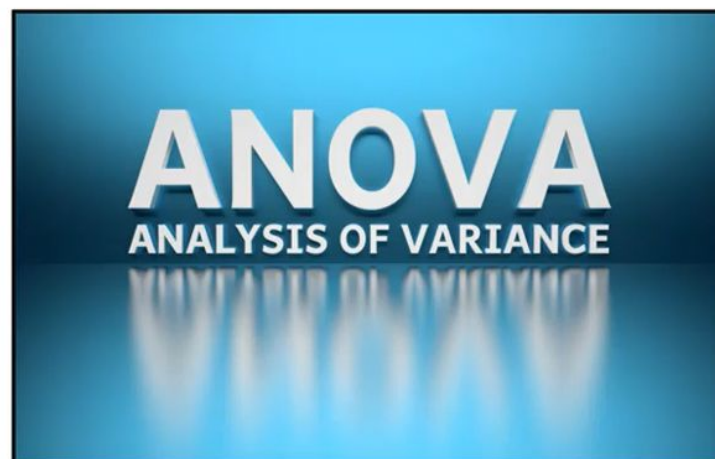
Test	Data Type	Relationship
Chi-Square Independence	Categorical (Counts)	Multiple groups/categories
2-Proportion Z-Test	Binary (Success/Fail)	Two independent groups
McNemar Test	Binary (Success/Fail)	Two dependent/matched groups

w3.4- ANOVA (6:31)

w3.4- ANOVA (6:31).mp4

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ANOVA



Earlier, we discussed an independent treatment t-test for two groups.



00:00:35 / 00:06:31

Assumptions for ANOVA

- Continuous dependent variable (DV)
- Independent variable (IV) is categorical with two or more independent groups
- Observations are independent
- Normality of dependent variable for each level of the independent variable
- No significant outliers
- Homogeneity of variance



ANOVA Example

Black	White	Indian	Asian	
69	51	26	18	
32	31	43	18	
37	60	15	30	
66	20	45	28	
60	47	30	30	
52	31	32	39	
	49	57	18	
	40	41	22	

Column	n	Mean	Std. Dev.	Std. Error
Black	6	52.66666	15.30577	6.248555
White	8	41.125	13.10874	4.634642
Indian	8	36.125	13.03224	4.607592
Asian	8	25.375	7.652030	2.705401

Source	DF	SS	MS	F-Stat	P-value
Columns	3	2671.741	890.58056	5.82817	0.0035
Error	26	3972.958	152.80609		
Total	29	6644.7			



w3.4- ANOVA (6:31).mp4

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Two Way ANOVA



00:04:39 / 00:06:31

60

Two Way ANOVA Example

		Food		
		A	B	C
X		-84	24	64
		-71	30	92
		-74	80	76
		-75	71	76
		-92	70	81
		-97	60	61
Age				
		86	65	-54
		69	86	-56
	Y	61	47	-97
		39	59	-40
		94	49	-65
		72	59	-88

Source	DF	SS	MS	F-Stat	P-value
Food	2	28704.67	14352.33	47.0534	<0.0001
Age	1	245.4444	245.4444	0.804677	0.3768
Food x Age	2	129654.2	64827.11	212.5324	<0.0001
Error	30	9150.667	305.0222		
Total	35	167755			



A **Two-Way ANOVA** (Analysis of Variance) is a statistical test used to determine how **two independent categorical variables** (factors) affect a continuous dependent variable.

While a One-Way ANOVA looks at one factor (e.g., "Freezer Brand"), a Two-Way ANOVA allows you to look at two factors simultaneously (e.g., "Freezer Brand" and "Storage Temperature") to see how they influence an outcome (e.g., "Ice Cream Hardness").

This module introduces **ANOVA (Analysis of Variance)**, a powerful statistical tool used when you need to compare the means of three or more groups simultaneously. While a T-test is limited to two groups, ANOVA allows for a broader investigation of multiple variables and their interactions.

1. One-Way ANOVA

One-Way ANOVA tests whether there are any statistically significant differences between the means of three or more independent groups.

- **The Hypotheses:**
 - **Null (H_0):** All group means are equal ($\mu_1 = \mu_2 = \mu_3 = \dots$).
 - **Alternative (H_1):** At least one group mean is different.
- **Post Hoc Analysis:** If you reject the null hypothesis, ANOVA doesn't tell you *which* group is different. You must perform a "Post Hoc" test (like **Tukey's HSD**) to identify the specific groups that differ.

Key Assumptions

To ensure valid results, six primary assumptions must be met:

1. **Continuous Dependent Variable:** (e.g., test scores, height, time).
2. **Categorical Independent Variable:** (e.g., ethnicity, treatment type).
3. **Independence of Observations:** No relationship between the subjects in each group.
4. **Normality:** The data in each group should be roughly normally distributed. (ANOVA is robust to slight violations here).
5. **No Significant Outliers:** Outliers can disproportionately pull the mean and skew results.
6. **Homogeneity of Variances:** The "spread" of data should be similar across all groups. This is typically verified using **Levene's Test**.

2. Example: Endurance Scores by Ethnic Group

In the transcript's example, endurance scores were compared across four ethnic groups ($\alpha = 0.05$):

- **Result:** The p-value was **0.0035**.
- **Decision:** Reject the Null ($0.0035 < 0.05$).
- **Post Hoc Finding:** Tukey's HSD revealed that **Black** and **Asian** groups were statistically different from each other, while other pairings did not show significant differences.

Group	Sample Size (n)	Mean Score	Std. Deviation
Group A (e.g., Black)	15	82.4	5.2
Group B (e.g., Asian)	15	74.1	4.8
Group C	15	78.5	5.1
Group D	15	79.2	4.9

3. Two-Way ANOVA

A Two-Way ANOVA expands the analysis by looking at **two independent variables** simultaneously.

- **Main Effects:** It tests the effect of each independent variable individually (e.g., Does **'Food'** affect the score? Does "Age" affect the score?).
- **Interaction Effect:** It tests if the effect of one variable depends on the other (e.g., Does a certain food only affect the score for specific age groups?).

Hypotheses in Two-Way ANOVA

1. **Variable A:** Is there a difference in means based on the first variable?
2. **Variable B:** Is there a difference in means based on the second variable?
3. **Interaction:** Is there an interaction between Variable A and Variable B?

4. Summary of ANOVA Types

Feature	One-Way ANOVA	Two-Way ANOVA
Independent Variables	1	2
Dependent Variable	1 (Continuous)	1 (Continuous)
Primary Goal	Compare group means	Compare means + check interactions
Example	Testing 3 different diets	Testing 3 diets across 3 age groups

w3.5-R-ANOVA and ANCOVA (10:42)

Module 3 - Lesson 5



R-ANOVA is used when there are multiple treatments on the same set of subjects.

Assumptions for R-ANOVA

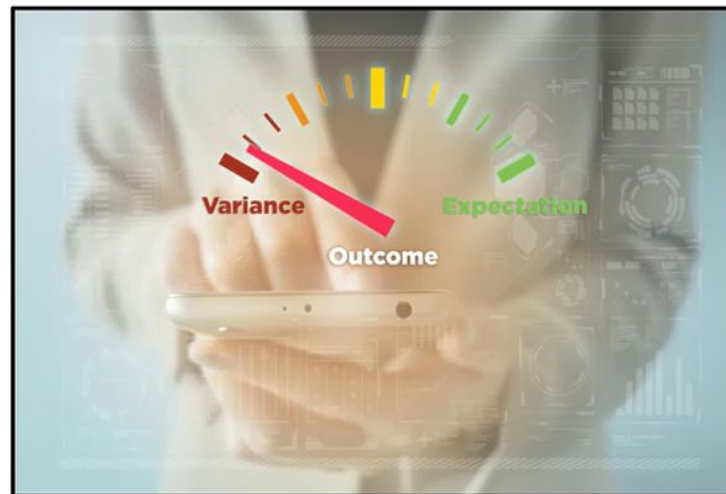
- Continuous dependent variable (DV)
- Within-subjects factor is categorical with three or more levels
- Normality of dependent variable for each level of the within-subjects factor
- No significant outliers
- Sphericity

There are numerous assumptions that  KENNESAW STATE UNIVERSITY must be met when running R-ANOVA.

Benefits of R-ANOVA



ANCOVA



In some situations, when using the ANOVA,

ANCOVA Application

Age Group 1		Age Group 2		Age Group 3	
GPA	Driving Ability	GPA	Driving Ability	GPA	Driving Ability
2.3	89	2.7	75	2.6	97
3.2	71	2.5	81	3.4	72
3.3	83	2.4	91	2.7	61
3.7	77	3.1	57	2.0	90
3.2	93	3.0	48	2.1	82
2.2	85	3.8	83	2.9	73

For example, we could use a one way ANCOVA to determine whether driving ability



ANCOVA stands for **Analysis of Covariance**. It is a statistical blend of an **ANOVA** and **Regression**.

While an ANOVA looks for differences in means between groups, an ANCOVA does the same thing but first "cleans" the data by removing the influence of an external variable called a **Covariate**.

1. What is a Covariate?

A covariate is a continuous variable that you suspect influences your dependent variable but isn't the main focus of your study. By including it, you reduce "noise" and make your test more powerful.

- **Example:** You want to test which **Teaching Method** (Factor) leads to higher **Test Scores** (Dependent Variable).
- **The Problem:** Some students are naturally smarter or had better grades before the class started.
- **The Solution:** You use their **IQ** or **Previous GPA** as a **Covariate**. ANCOVA mathematically adjusts the test scores as if every student started with the same IQ.

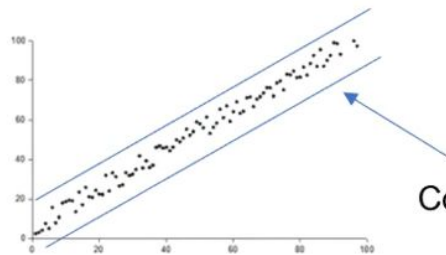
Assumptions for ANCOVA

- Continuous dependent variable (DV)
- Independent variable (IV) is categorical with two or more independent groups
- Continuous covariate variable
- Observations are independent



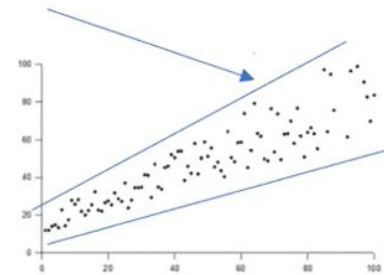
Assumptions for ANCOVA (cont.)

- Normality of dependent variable for each grouping of the independent variable
- Homoscedasticity



Constant error variance

Increasing error variance



First, normality of
the dependent variable for



Assumptions for ANCOVA (cont.)

- Homogeneity of variance
- No significant outliers
- Covariate variable is linearly related to the dependent variable at each level of the independent variable
- Homogeneity of regression slopes

Additionally, covariate variables
should be linearly related to



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Benefits of ANCOVA



Identifying and
controlling known extraneous variables or



Example for ANCOVA

CV= Covariate Variable

Black		White		Indian		Asian	
CV	DV	CV	DV	CV	DV	CV	DV
51	69	47	51	46	26	34	18
42	32	24	31	50	43	35	18
55	37	38	60	28	15	46	30
56	66	28	20	58	45	22	28
52	60	62	47	24	30	45	30
50	52	47	31	50	32	53	39
		60	49	53	57	33	18
		38	40	24	41	34	22

Let's go back to the ANOVA example and introduce height as a covariate variable.



Example for ANCOVA (cont.)

ANCOVA Results (k=4)						
Source	SS	df	MS	F	P	
adjusted means	1291.547	3	430.5156	3.628898	0.02663	
adjusted error	2965.884	25	118.6354			
adjusted total	4257.431	28				
Test for Homogeneity of Regressions						
Source	SS	df	MS	F	P	
between regressions	144.0014	3	48.00045	0.425252	0.736591	
remainder	2821.882	25	112.8753			
Correlation Between CV and DV			0.599395			



This module concludes our deep dive into advanced ANOVA variants, focusing on **Repeated Measures ANOVA (R-ANOVA)** and **Analysis of Covariance (ANCOVA)**. These tools provide Six Sigma practitioners with the ability to control for subject fatigue and outside "noise" variables.

In R, an **ANOVA** (Analysis of Variance) is a statistical procedure used to determine if there are significant differences between the means of three or more independent groups. While a t-test compares two groups, ANOVA handles multiple groups simultaneously, preventing the "inflation" of error rates that happens when running many t-tests. 

1. What does it actually do?

The core "job" of an ANOVA is to partition the total variance in your data into two distinct parts:



1. **Between-Group Variance:** How much the averages of the groups differ from each other.
2. **Within-Group Variance (Error):** How much the individual data points within each group spread out from their own group mean.

The test calculates an **F-statistic**. If the variation *between* groups is much larger than the variation *within* groups, the F-statistic will be high, and the p-value will be low (below **0.05**), suggesting that the groups are not all the same. 

1. Repeated Measures ANOVA (R-ANOVA)

R-ANOVA is used when you test the **same subjects** under multiple conditions. Think of it as a 2-sample paired T-test, but for three or more treatments.

- **Benefit:** Since the same people are used for every test, you eliminate person-to-person variation, making the test much more sensitive **to treatment effects with fewer samples.**
- **The Risk of Fatigue:** Subjecting a participant to multiple treatments can lead to "order effects" or fatigue, which may bias your results.

The Assumption of Sphericity

Unique to R-ANOVA, **Sphericity** is the requirement that the variances of the differences between all combinations of treatments are equal.

- **Violation:** If sphericity is violated (which is common), the risk of a Type I error (false positive) increases.
- **Correction:** Use conservative adjustment factors like **Greenhouse-Geisser** or **Huynh-Feldt** to adjust the degrees of freedom.

2. ANCOVA (Analysis of Covariance)

ANCOVA is a "hybrid" test that combines ANOVA with linear regression. It allows you to "subtract" the influence of an outside variable (the **Covariate**) to see the true effect of your independent variable.

- **The Covariate:** A continuous variable that is linearly related to your outcome.
 - *Example:* Testing **Driving Ability** (DV) based on **Age Group** (IV) while controlling for **GPA** (Covariate).
- **Purpose:** By removing the "noise" caused by the GPA, you get a clearer, more deterministic view of how Age actually affects driving.

Critical Assumptions for ANCOVA

Beyond standard ANOVA requirements, ANCOVA adds:

1. **Linearity:** The covariate and dependent variable must have a linear relationship.
2. **Homogeneity of Regression Slopes:** The relationship between the covariate and the dependent variable must be the same for all groups (visually, the regression lines should be **parallel**).
3. **Homoscedasticity:** The error variance should remain constant across all levels of the independent variable.

3. ANOVA vs. ANCOVA: A Case Study

The transcript illustrates a comparison of endurance scores with and without a covariate (Height):

Feature	One-Way ANOVA	ANCOVA (with Height)
P-Value	0.0035	0.0266
Confidence	Very High	Moderately High
Insight	At least one group differs.	A difference exists, but some was due to height.

Observation: While the result remained significant, the p-value increased after adding the covariate. This means some of the "difference" we originally saw in the groups was actually explained by the height of the participants, not just their group membership.

Summary Table: Which Test to Use?

Scenario	Recommended Test
Same subjects tested at 3 different time intervals	Repeated Measures ANOVA
Different groups compared while ignoring outside factors	One-Way ANOVA
Different groups compared while "canceling out" a noise variable	ANCOVA

Repeated Measures ANOVA is used when you take multiple measurements of the same dependent variable from the same subjects under different conditions or at different time points. 

Unlike a standard ANOVA, which compares different groups of people, Repeated Measures ANOVA compares the **same group** to itself across various "levels."

1. Common Usage Scenarios

- **Changes Over Time (Longitudinal Studies):** Measuring the same participants at multiple intervals to see how they change. 
 - *Example:* Measuring patient blood pressure before treatment, 1 month after, and 6 months after.
- **Response to Different Stimuli (Within-Subjects):** Testing how one person reacts to different conditions.
 - *Example:* Testing a person's reaction time to a red light, a green light, and a blue light.
- **Dose-Response Trials:** Observing how a subject reacts to increasing levels of a medication or caffeine.

